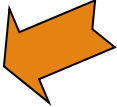


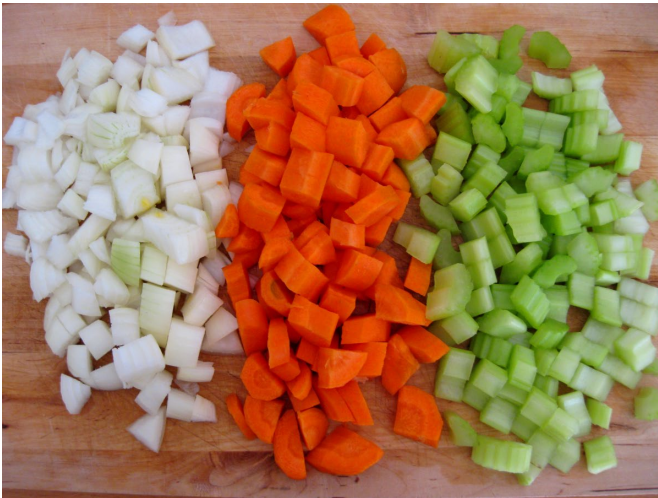
Pattern Mining: Concepts and Typical Methods

- ☐ Basic Concepts 
- ☐ Frequent Itemset Mining Methods
- ☐ Which Patterns Are Interesting?—Pattern Evaluation Methods
- ☐ Mining Various Kinds of Patterns
- ☐ Summary

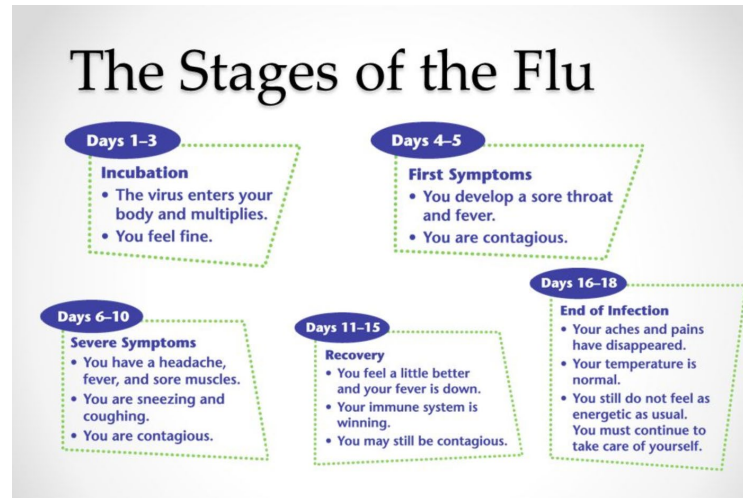
What Are Patterns?

❑ What are patterns?

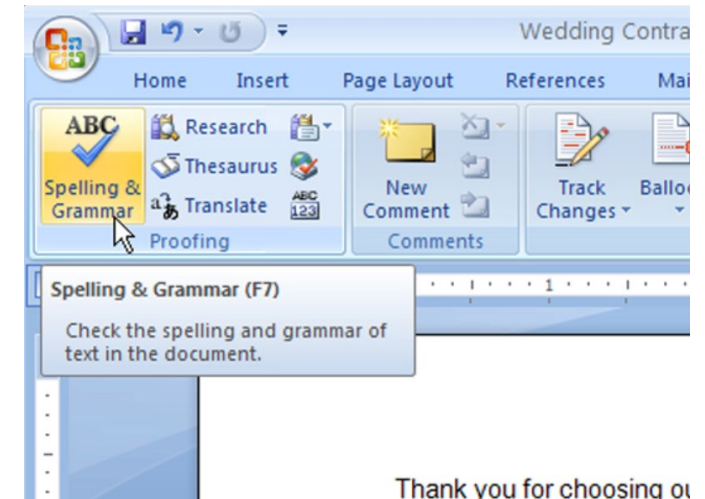
- ❑ **Patterns**: A set of items, subsequences, or substructures that occur frequently together (or strongly correlated) in a data set
- ❑ Patterns represent **intrinsic** and **important properties** of datasets



Frequent item set



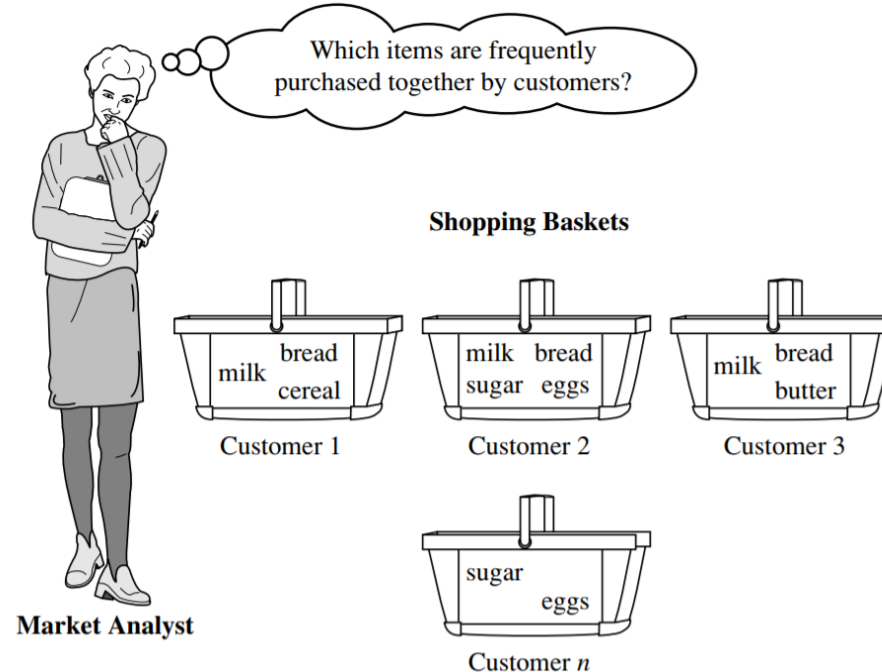
Frequent sequences



Frequent structures

What Is Pattern Discovery?

- ❑ **Pattern discovery:** Uncovering patterns from massive data sets
- ❑ It can answer questions such as:
 - ❑ What products were often purchased together?
 - ❑ What are the subsequent purchases after buying an iPad?



Pattern Discovery: Why Is It Important?

- ❑ **Foundation** for many essential data mining tasks
 - ❑ Association, correlation, and causality analysis
 - ❑ Mining **sequential**, structural (e.g., sub-graph) patterns
 - ❑ **Classification**: Discriminative pattern-based analysis
 - ❑ **Cluster** analysis: Pattern-based subspace clustering
- ❑ **Broad applications**
 - ❑ Market basket analysis, cross-marketing, catalog design, sale campaign analysis, Web log analysis, biological sequence analysis
 - ❑ Many types of data: spatiotemporal, multimedia, time-series, and stream data